

Ex. 2: Implementing TDMA Slot Allocation in GSM

Code:

```
clc;
clear;
close all;
Fs = 1000;
T = 1;
N = 3;
t = 0:1/Fs:T-1/Fs;
user1 = sin(2*pi*5*t);
user2 = sin(2*pi*10*t);
user3 = sin(2*pi*15*t);
slot_length = floor(length(t)/N);
tdma_signal = zeros(size(t));
tdma_signal(1:slot_length) = user1(1:slot_length);
tdma_signal(slot_length+1:2*slot_length) = ...
user2(slot_length+1:2*slot_length);
tdma_signal(2*slot_length+1:end) = ...
user3(2*slot_length+1:end);
figure;
subplot(4,1,1)
plot(t,user1,'b','LineWidth',1.5);
title('User 1 Signal (5 Hz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,2)
plot(t,user2,'r','LineWidth',1.5);
title('User 2 Signal (10 Hz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,3)
plot(t,user3,'g','LineWidth',1.5);
title('User 3 Signal (15 Hz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,4)
```

```

plot(t,tdma_signal,'k','LineWidth',1.5);
title('Combined TDMA Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```

